

Yuesheng Zhang

PhD student in industrial biotechnology
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EDUCATION

The Novo Nordisk Foundation Center for Biosustainability at DTU **2024.03 - present**
(Current name: The Novo Nordisk Foundation Biotechnology Research Institute for the Green Transition (BRIGHT))

PhD student | Major: Industrial Biotechnology

School of Biotechnology, Jiangnan University, China **2020.09 - 2023.06**

Master of Engineering | Major: Biology and Medicine (Bioengineering)

School of Pharmaceutical Science, Jiangnan University, Wuxi, China **2016.09 - 2020.06**

Bachelor of Engineering | Major: Pharmaceutical Engineering

PROFESSIONAL EXPERIENCE

Shenzhen International Graduate School, Tsinghua University, China **2023.07 - 2024.01**

Research Assistant | Department: Institute of Biopharmaceutical and Health Engineering

PUBLICATIONS

(#: first author, *: corresponding author)

[1] Xiongwen Li, Zhetao Guo, Yi Li, Zhu Liang, Ziyi Liu, Ke Wu, Jiahao Luo, **Yuesheng Zhang**, Lizheng Liu, Manda Sun, Yuanyuan Huang, Yu Chen, Tao Yu, JensNielsen, Hongting Tang, Feiran Li. Leveraging large language models for metabolic engineering design. *Trends in Biotechnology*. 2026.

[2] **Yuesheng Zhang**, Douglas B. Kell, Irina Borodina. Engineering and characterization of small-molecule transporters for cell factories. *Trends in Biotechnology*. 2025.

[3] Jia-Yu Jiang#, Dai Xue#, Jin-Song Gong, Qin-Xin Zheng, **Yue-Sheng Zhang**, Chang Su, Zheng-Hong Xu, Jin-Song Shi. A comprehensive review on microbial hyaluronan-degrading enzymes: from virulence factors to biotechnological tools. *Bioresources and Bioprocessing*. 2024, 312:120809.

[4] **Yue-Sheng Zhang**#, Jin-Song Gong*, Jia-Yu Jiang, Zheng-Hong Xu and Jin-Song Shi*. Engineering protein translocation and unfolded protein response enhanced human PH-20 secretion in *Pichia pastoris*. *Appl Microbiol Biotechnol*. 2024, 108: 54.

[5] **Yue-Sheng Zhang**#, Jia-Yu Jiang, Jin-Song Gong*, Chang Su, Heng Li, Chuan-Li Kang, Lei Liu, Zheng-Hong Xu and Jin-Song Shi*. High-level expression and characterization of a highly active hyaluronate lyase HylC with significant potential in hyaluronan oligosaccharide preparation. *Journal of Biotechnology*. 2023, 366: 35-45.

[6] Zhi-Yuan Yao#, Jin-Song Gong*, Yu-Ru Liu, Jia-Yu Jiang, **Yue-Sheng Zhang**, Chang Su, Heng Li, Chuan-Li Kang, Lei Liu, Zheng-Hong Xu, Jin-Song Shi*. Genetic variation reveals the enhanced microbial hyaluronan biosynthesis via atmospheric and room temperature plasma. *Carbohydrate Polymers*. 2023, 312:120809.

[7] **Yue-Sheng Zhang**#, Jin-Song Gong*, Zhi-Yuan Yao, Jia-Yu Jiang, Chang Su, Heng Li, Chuan-Li Kang, Lei Liu, Zheng-Hong Xu and Jin-Song Shi. Insights into the source, mechanism and biotechnological applications of hyaluronidases. *Biotechnology Advances*. 2022, 60:108018.

PATENTS

[1] Jin-Song Shi, Yue-Sheng Zhang, Jin-Song Gong, Zheng-Hong Xu, Chang Su, Heng Li, Jia-Yu Jiang. A recombinant *Pichia pastoris* expressing a human hyaluronidase and its applications. CN (Patent) 202311030579.1, filed August 16, 2023

[2] Jin-Song Gong, Yue-Sheng Zhang, Jin-Song Shi, Zheng-Hong Xu, Heng Li, Chang Su, Jia-Yu Jiang, Shao Jie Lian, Chuan-Li Kang, Lei Liu, De-Qiang Zheng. Sequence, recombinant strain and application of a hyaluronate lyase. CN (Patent) 202310217653.4, filed March 8, 2023.

PRESENTATIONS

[1] Unveiling transporter substrate range by untargeted metabolomics. Poster. Metabolic Engineering 16, Copenhagen, 2025.

RESEARCH EXPERIENCE

Sustainable production of Single-cell protein for future foods 2025 -

- Food-grade fungi was engineered to assimilate alternative carbon sources from CO₂ conversion and waste streams

Metabolic engineering of yeast to produce plant natural products 2024 -

- Engineering *S. cerevisiae* and *Y. lipolytica* to produce food colorant from plant
- Pathway engineering and mining promiscuous reactions to eliminate byproducts

Membrane transporter characterization and engineering 2024 -

- A *Xenopus* oocyte-based method was established to characterize promiscuous activity of transporters
- Using Adaptive evolution and reverse engineering to achieve yeast growth on novel substrates facilitated by new transport activity

Leverage mechanistic model and deep learning to study protein production 2023 - 2024

- A mathematical metabolic network model for *Pichia pastoris* was generated by a semi-automated workflow
- The model precisely predicts the optimal growth rate for protein production, and guides strain engineering

Industrial application of polysaccharide degrading enzyme and its producing strain 2020 - 2023

- High level production of hyaluronidase was achieved in 5-L fermenter
- Scale-up production in 20-L pilot plant fermenter (Company collaboration project)

Strain engineering for enhanced secretory protein expression in microbial hosts 2020 - 2023

- PH-20, a human hyaluronidase was expressed using *P. pastoris* system
- The protein secretion pathway was engineered to improve PH-20 secretion by 2 times

SKILLS (Selected)

Biomanufacturing

- 5 years of experience in yeast metabolic engineering
(*Y. lipolytica*, *S. cerevisiae*, *P. pastoris*)

Scientific research

- Proved expertise in summarizing state-of-the-art and writing high-quality literature reviews
- Independent design of scientific research projects
- Communicating research result with

- Protein expression and purification
(*P. pastoris*, *Xenopus oocyte*, *E. coli*)
- A background combining wet lab and dry lab
- Experiences in industry collaboration and scaling up

Office & Collaboration

- Leading 4 collaborative research projects in an international environment
- Supervised 3 bachelor thesis and 1 master thesis
- Proficient in oral and written English